

Urban buoyancy-driven air flow and modelling method: A critical review

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ARTICLE INFO

Keywords:

Thermal buoyancy
Urban ventilation
Urban heat island
CFD
Surface temperature model

ABSTRACT

The evaluation of buoyancy effect on urban ventilation, especially on low wind conditions, is important for the extreme urban heat islands to improve urban climate resilience. To provide suggestions for reliable climate simulation study in the future and support the development of city-scale outdoor ventilation models, this study conducted a critical literature review on buoyancy effects on outdoor natural ventilation. First, on the neighborhood scale, this study summarized computational fluid dynamics (CFD) simulations on urban ventilation with buoyancy effects to provide suggestions on improving simulation accuracy. Suggestions on the computational mesh, turbulence models, and wall functions, as well as the realistic wall surface temperature models coupling, are provided. Furthermore, the newly emerged non-isothermal wind tunnel measurements are summarized to support future validation studies. We highlight the importance of using Large-Eddy Simulation model under weak wind and intense wind-buoyancy interaction scenarios. The Building Energy Model (BEM) is recommended to calculate the surface temperature and couple it into CFD models. Secondly, on the urban scale, in order to support the parameterization of buoyancy effect on air exchange velocity, correlations between air exchange velocity and non-dimensional parameters are explored based on existing simulation results. A linear correlation is found between the street canyon buoyancy parameter and air exchange velocity improvement. This review suggests that the current database of air exchange velocity is sparse and limited at a narrow range of buoyancy intensity and urban densities, and thus emphasizes the need for future studies on strong buoyancy and weak wind conditions.

1. Introduction

The densely packed buildings reduce the wind and block the radiation, leading to the urban heat islands (UHI) effect, which means the urban air/surface temperature is higher than its surrounding rural area. For example, a maximum urban heat island of 7 °C was observed in Singapore [1]. The high ambient air temperature increases energy consumption for cooling, and is responsible for higher mortality and morbidity [2]. To build a sustainable and resilient urban environment, a good understanding of urban heat dispersion is needed. Urban natural ventilation is regarded as a promising strategy to alleviate the urban heat islands (UHI) [3,4].

Urban natural ventilation, also known as “urban outdoor ventilation” or “city breathability”, is a concept to assess the removal potential of pollutants, heat and moisture out of the urban canopy layer (UCL) [5–7]. Here, the UCL refers to the space below the rooftops of urban buildings [8]. The environment in the UCL has a direct impact on residents’ health and comfort. Many indices were developed by previous studies to

quantify the urban ventilation, such as the exchange velocity [9], purging flow rate [10], residence time [11], mean age of air [7], and etc., [5]. With these indices, many factors were found to influence the urban natural ventilation rate, including wind speed [12], wind direction [13], building porosity [7,14], building form [15], building height variation [16], building layout [17], and traffic-induced turbulence [18, 19]. Among these factors, buoyancy is a critical parameter for urban ventilation, especially under an intense UHI condition with a light incoming wind [20].

The buoyancy force is generated by hot urban surfaces, which are mainly heated by solar radiation at daytime, as well as and man-made waste heat. Firstly, urban building arrays trap the reflected solar radiation [21] and also man-made waste heat [22]. Secondly, the urban surfaces lack moisture evaporation [23]. Thirdly, the construction materials are of low heat capacity and easily heated up to high temperature. For example, the asphalt street in Greece could be heated up to 60 °C in summertime [24]. At nighttime, the cooling of urban surfaces is hindered by blocking the longwave radiation [25,26]. The heated urban

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<https://doi.org/10.1016/j.buildenv.2021.108708>

Received 1 November 2021; Received in revised form 8 December 2021; Accepted 20 December 2021

Available online 24 December 2021

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surfaces have a significant impact on the urban flow pattern [27,28]. The correlation between high-temperature surfaces and street air temperature is not straightforward. On one hand, heated urban surfaces will warm the air. On the other hand, the air exchange rate can be enhanced by the buoyancy effect [29].

The urban airflow can be completely changed by buoyancy. Under strong buoyancy conditions, the thermal plumes could penetrate and overcome the shear layer flow [30]. The buoyancy may also interact with the complex urban geometry. For example, the air exchange capacity in steep canyon was found much lower than in the flat canyon [30]. In that case, the downward flush flow along the windward façade of the tall building suppresses the upward buoyant plumes, lowering the air exchange. However, the building height variation enhances the exchange velocity when ignoring the buoyancy effect [16]. To accurately predict the UCL thermal environment, it is essential to understand the buoyancy effect and its interaction with ambient wind and urban geometry.

Urban airflow is essentially multiple-scale. Britter and Hanna [31] divided the urban airflow into four scales: regional (~100 or 200 km), city scale (~10 or 20 km), neighborhood scale (~1 or 2 km), and street scale (less than 100–200 m). For the non-isothermal urban natural ventilation with buoyancy effects, most studies focus on the street scale [32–36] and neighborhood scale [10,37–39]. A few include the city scale [28,40].

On the street/neighborhood scale, the urban geometries, including the buildings, streets, trees, etc., are explicitly included to investigate the detailed airflow and heat dispersion within and above the urban canopy layer [41]. Field measurements, wind tunnel experiments, and Computational Fluid Dynamics (CFD) are three major approaches to obtain airflow fields. Due to the high cost of field measurements and wind tunnel experiments, they work as a benchmark to validate CFD models. The CFD models have been widely used to simulate airflow in urban areas and explore the thermal effect on the pollutant removal rate and air exchange rate [42]. At the street/neighborhood scale, the resolution of the CFD simulation could be less than 1 m, which provides detailed flow fields and temperature fields [43]. The modelling results can help urban planners to improve their design to cope with the UHI under extreme heat conditions.

Urban flow prediction under non-isothermal conditions are significantly influenced by the surface temperature model and turbulence model. First, the urban surface temperature is influenced by the coupling of solar radiation, long wave radiation, heat conduction and storage at building materials and urban soil, evaporation, and the convection driven by urban flow [44]. Various surface temperature models with different simplification methods were used in the previous studies [20,45], and more importantly, they cause conflict conclusions between studies. Secondly, the performance of turbulence models has been widely discussed under the isothermal conditions, and shows a large difference between models [46,47]. However, studies focusing on non-isothermal conditions are relatively scarce. The accuracy of many previous simulations is limited due to the unsuitable turbulence model and surface temperature model. As a result, their application in real urban design practice is limited. Recent studies have found many new understandings that can improve the simulation accuracy, while unfortunately these new understandings have not been widely applied in the existing studies that we reviewed in this paper. To improve the applicability of simulation results, there is an urgent need to systematically review and summarize these studies.

On the city scale, the application of CFD model is limited due to the high computational cost. Semi-empirical models, in which urban geometrical parameters are applied to estimate climate factors, are more feasible than CFD [40]. The popular models at the city scales include the Urban Weather generators (UWG) [48], SIRANE [49], and the Multiple layer canopy model [40]. The estimation of exchange velocity, which is the average velocity of mass transfer into or out of urban canopy [9], is critical for semi-empirical models. There is a comprehensive database on

the mass and momentum exchange rate for urban geometry [9,50,51]. The buoyancy plays a dominant role in the weak wind conditions, which correspond to the extremely high urban heat island and pollutant island events [29,52]. This weather condition can be found during heat waves [53]. Additionally, UHI may interact with heat waves and prolong and intensify hot conditions [54–57], putting severe heat stress on urban residents. To cope with this challenge, predicting models for local air temperature under strong buoyancy condition are urgently needed. However, there is a lack of quantification of the buoyancy effect on air exchange velocity. This limitation makes the current city-scale models unable to deal with strong buoyancy conditions, which corresponds to the extreme UHI scenarios and is important to the urban climate resilience analysis.

In summary, thermal buoyancy is important for urban natural ventilation, especially during heat waves with high UHI intensity and weak wind conditions. This study conducted a multi-scale literature review. First, to help future street/neighborhood scale simulation studies improve modelling accuracy, this study critically reviewed previous studies to provide suggestions for turbulence modelling and surface temperature modelling. Secondly, for the purpose of developing city-scale models, this study provides a systematic review of the buoyancy effect for air exchange velocity and explores its parameterization based on the existing simulation results.

The mechanism of buoyancy effect on urban air flow is firstly discussed in Section 2. Section 3 and Section 4 respectively summarize the turbulence models and surface temperature models suitable for urban air flow modelling with buoyancy effect. The buoyancy coefficients calculated from previous studies are presented for the parameterization in Section 5. The conclusions and future directions are provided in Section 6.

2. Governing parameters and wind-buoyancy interactions

The urban ventilation flow pattern depends on the relative strength, i.e., a ratio of ambient wind to buoyancy, because the synoptic wind and thermal buoyancy always exist simultaneously in the real world. Therefore, it is essential to carry out a dimensional analysis to find out the impact of relative strength, e.g., how strong the buoyancy is strong enough to affect the urban ventilation flow pattern. The dimensionless numbers to evaluate the buoyancy effect are summarized. Then, the street canyon airflow that is driven by both ambient wind and local buoyancy is discussed. Following that, the street canyon airflow that is exclusively driven by local buoyancy is discussed.

2.1. Dimensionless numbers to evaluate the buoyancy effect

Several parameters were proposed to quantify the relative strength between ambient wind and buoyancy, including the dimensionless numbers in fluid mechanics: Froude number (Fr), and Richardson number (Ri) [58]. These two parameters can be regarded as equivalent as they are the reciprocal to each other, i.e., $Ri = 1/Fr$. Large absolute value of Ri number indicates a strong buoyancy effect and negative value indicates positive buoyancy, and naturally, the small value of Fr number shows strong buoyancy effect. The definition of Richardson number was frequently written as $Ri = Gr/Re^2$, where Gr is the Grashof number and Re is the Reynolds number. Their definitions are as follows,

$$Re = \frac{u_0 H}{\nu} \quad (1)$$

$$Gr = \frac{g \beta H^3 (T_w - T_\infty)}{\nu^2} \quad (2)$$

where u_0 : ambient wind speed, g : gravitational acceleration, H : characteristic length (e.g., building heights), β : thermal expansion coefficient, T_w : surface temperature, T_∞ : ambient air temperature, ν : air kinematic viscosity. For idealized building configuration and uniform

surface temperature, the selection of length scale H and temperature differences ($T_w - T_\infty$) is straightforward. However, it becomes complex for cases with multiple building heights and non-uniform surface temperature, which could lead to large variations in the estimated Ri or Gr . The cross-comparison between studies should be cautious when selecting the length scale and temperature difference. For example, Zhao et al. [30] selected the height of lower building as the length scale for a step-up street canyon. In non-uniform wall heating condition, Nazarian and Kleissl [59] proposed a horizontal Richardson numbers Ri_h as:

$$Ri_h = \frac{g\beta H(\bar{T}_L - \bar{T}_W)}{U_b^2} \frac{H}{W} \quad (3)$$

Where $\bar{T}_w$ and $\bar{T}_L$ are the averaged surface temperature on windward and leeward walls, respectively.

The Richardson number used in this study is the bulk Richardson number [60]. It should be noted that Richardson number has other forms, including the gradient Richardson number (Ri_g) and flux Richardson number (Ri_f), which are used to evaluate local instability of a thin atmospheric layer [61,62]. The gradient Richardson number is generally defined as:

$$Ri_g = \frac{g}{T_v} \frac{\partial \theta_v / \partial z}{(\partial U / \partial z)^2 + (\partial V / \partial z)^2}, \quad (4)$$

where g is the gravity, T_v the absolute virtual temperature, θ_v the virtual potential temperature, z the depth of the layer, U and V the horizontal wind components [63]. The Ri_f is defined as the ratio of buoyancy destruction to shear production of turbulent kinetic energy [62]. The Ri_f is related to Ri_g by the turbulent Prandtl number ($Pr_t = Ri_g / Ri_f$). In the following sections, the Richardson number refers only to the bulk Richardson number and should not be confused with gradient Richardson number and flux Richardson number.

The airflow pattern at the urban canopy layer is found strongly related to these dimensionless parameters. For example, the wind tunnel experiment with a heated cubic block shows the reattachment lengths behind the building decreased with Ri when $Ri < -0.2$, due to strong buoyancy effect, and unchanged when $Ri > -0.2$ [58]. A CFD simulation of street canyon shows that the single vortex flow was transferred into two vortices flow when $Ri < -2$ under heated windward wall conditions [64]. For a heated ground surface, two counter-rotating flow structures appeared when $Ri < -4$. The simulation of detected one or two stable counter-rotating vortices at high Fr numbers ($\approx 10^1$) and lower vortex

dominated flow at low Fr numbers ($\approx 10^{-1}$) [65].

The urban geometrical configurations and heating position also change the urban airflow. Apart from the horizontal Richardson numbers, another buoyancy parameter $B = (g\alpha\Delta TH) / (u_0^2[1 + (H/W)^2])$ also couples Ri with urban geometrical characteristics, where u_0 is the ambient wind velocity, and ΔT is the temperature difference between upwind wall and downwind wall [66]. The derivation of B is based on canyons that are occupied by a single circulation cell under an assisting condition, i.e., the wind-driven and buoyancy-driven airflow are of the same direction. Therefore, this parameter is only applicable to aspect ratio that ranges from 0.3 to 1.2. In this case, the buoyancy has a larger impact in wider street canyons. Otherwise, multiple vortices may exist in the street canyon. In that case, the buoyancy parameter B is no longer applicable. The buoyancy is likely to have a stronger influence in a deeper street canyon as the street canyon airflow weakens. For example, the horizontal Richardson number Ri_h shows stronger buoyancy effect in deep street canyon [59].

A criterion value $B_c \approx 0.05$ was proposed to show when buoyancy can be ignored [66]. When $B < B_c$, the wind-driven circulation prevails in-canyon flow. When $B > B_c$, the buoyancy driven flow cannot be ignored and at $B \gg B_c$, the flow is dominated by the buoyancy driven flow. Fig. 1 shows the correlation between B and u_0 with different geometry densities, i.e., $H/W = 0.7, 1.0$, and 1.2 . The weather station in Singapore shows that wind speeds of $0 \sim 6$ m/s are of high occurrence frequency [67]. As the buoyancy parameter B assumes a steady state flow, Fig. 1 only shows the B values for wind speed lower than 6 m/s. For a temperature difference of 3K, B exceeds the criterion value at most wind speeds. For the annual mean wind speed ($=2.65$ m/s in Singapore), B exceeds the criterion value with a small temperature difference of 1 K. This indicates the buoyancy cannot be ignored in most cases, especially for the low wind and large temperature difference conditions.

The thermal impact on street canyon flow is also influenced by the length scale. Zhao et al. [61] suggested that thermal effect becomes less significant with increasing building height. They calculated the ratio between the street canyon width and thermal boundary layer thickness, and found it increases from 10 to 30 when the building height increases from 0.1 m (wind tunnel building scale) to 10 m (full scale). In other words, the thermal boundary layer occupies 1/10 of the street canyon in reduced scale measurement and occupies 1/30 in full scale. Both Richardson number and buoyancy parameter B ignore this scale problem. Particular caution is needed when extending reduced scale results to full scale.

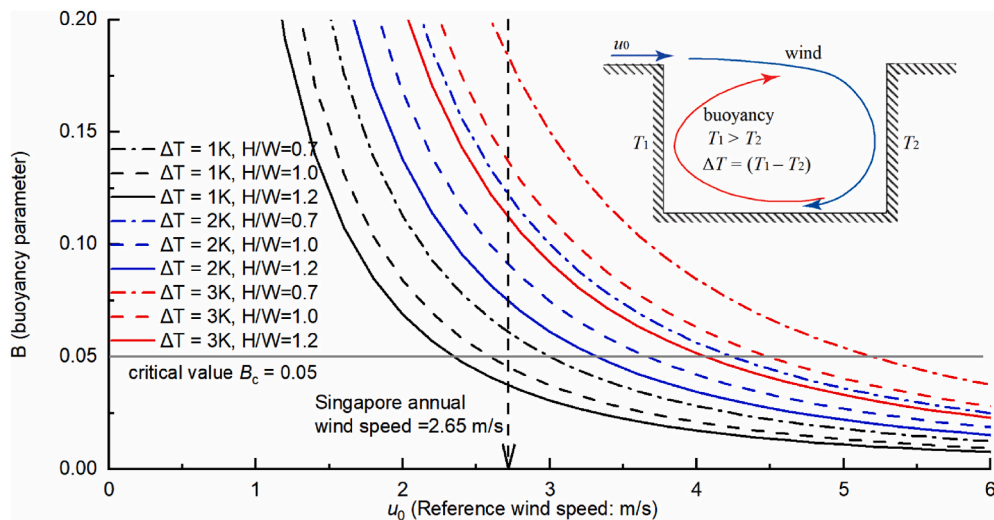


Fig. 1. Distribution of buoyancy parameter (B) at a typical urban configuration with $H/W = 0.7, 1.0$, and 1.2 (with a constant height of $H = 20$ m) under different wind speeds and urban surface temperature increments. The criterion value B_c is plotted as grey horizontal line. The area above it indicates a non-negligible buoyancy influence.

2.2. Interaction between buoyancy and wind-driven flow

Table 1 summarized 2D CFD studies on airflow and ventilation performance with buoyancy effect under a relatively strong ambient wind. The interaction between ambient wind and local buoyancy is the main driving factor of urban airflow. In the 2D simulations, the street canyon is regarded as the basic element of an urban area. The impacts of aspect ratio, heating position, and Ri number on the ventilation performance are the main topics of 2D studies. The height-width aspect ratio (H/W) is a critical geometrical feature for 2D street canyon. The deep street canyon with high aspect ratio causes weak wind inside the street canyon under isothermal condition [68]. As a result, the effect of buoyancy on airflow is more significant in the deep street canyon [69]. The original single vortex flow driven by ambient wind can be divided into a multi-vortices low regime [42]. Due to the low transport rate between the vortex cells, the pollutant dilution rate of street canyon significantly decreased. A wide range of aspect ratios are used in these studies, ranging from 0.5 to 5.0. These studies show a variety of flow patterns inside the street canyon under the wind-buoyancy driving condition. The resultant vortex number inside the street canyon ranges from 1–5, depending on the relative strength of buoyancy and ambient wind, which is generally represented by Ri number. The vortex number depends on both the aspect ratio and Ri number and their interactions. However, there is a lack of simple expressions to link the flow pattern to the aspect ratio and Ri number. Intensive simulations and measurements are still needed if the geometry or wind conditions changes.

Multiple indices were also used in previous studies to quantify the buoyancy effect on air exchange, such as the residue concentration ratio [11], air exchange rate (AER) [70], exchange velocity [27], pollutant retention time [71]. A strong correlation was found between the exchange rate and surface temperature, e.g., a second-order polynomial relationship between AER and Ri for the bottom heating conditions [70, 71]. For the façade heating condition, a linear relationship was found between the exchange velocity and Ri [27]. These studies show the possibility of using simple formulas to estimate the buoyancy effect without conducting new measurements or simulations.

Apart from urban geometries, the urban airflow pattern is also influenced by the position of heat sources. The interaction between ambient wind and buoyancy generates a complex airflow at the air volume below the roof level. Two mechanisms of the heated wall influence: the assisting condition and opposing condition, where the buoyancy-driven flow is of the same direction or opposite direction to the wind-driven flow, shown as Fig. 2 [28]. Under the opposing condition, multiple vortices flow can form in street canyons [27]. Ground heating was widely found of positive effect on the airflow speed inside street canyon [72–74]. The vortex number inside a street canyon can be very different under the influence of buoyancy [69,75,76]. These studies show that the urban ventilation performance could be enhanced by the buoyancy effect.

Table 2 summarized the 3D CFD simulations. Compared to a 2D street canyon, the 3D building arrays represent a more realistic urban configuration. The 3D airflow features such as vortices at street ends are

Table 1

CFD studies on influences of heated wall on airflow and ventilation within street canyon. NH (Not heated) GH (ground heated) AH (All wall heated) DH (downwind wall heated) UH (upwind wall heated).

References	Turbulence models	Heated surface	H/W	Vortex number	Ventilation indices	Ri
[42]	Standard $k-\epsilon$ model; Standard wall function	NH, GH, UH, DH	1.12	1–2	Cloud passing time	–0.13
[11]	Standard $k-\epsilon$ model; Standard wall function	NH, GH, UH, DH	0.5–3.5	1–3	Residue concentration ratio	–0.755 ~ –0.363
[69]	Standard $k-\epsilon$ model; Standard wall function	GH	0.6–3.6	1–3	Flow velocity	–1.81–0
[64]	Standard $k-\epsilon$ model; Standard wall function	NH, GH, UH, DH	1.0	1–2	Pollutant concentration	–4.75
[77]	Standard $k-\epsilon$ model; Standard wall function	NH, GH, UH, DH	1.0	1–2	Pollutant concentration	–39.04 ~ –1.1
[70]	standard, RNG, and realizable $k-\epsilon$ model; Standard wall function	GH	1.0	N/A	Air exchange rate	–34–0
[78]	RNG $k-\epsilon$ model; Standard wall function	GH, UH, DH, AH	0.1–2	1–3	Pollutant concentration	–6.6
[79]	RNG $k-\epsilon$ model; Standard wall function	GH	1.0	1	Pollutant concentration	Not given
[80]	RNG $k-\epsilon$ model; Standard wall function	GH	0.5–2	1–2	Air exchange rate	–5–0
[65]	Standard $k-\epsilon$ model; Standard wall function	DH	0.7–1.5	1–2	Thermal conductance	–16.7 ~ –0.09
[81]	LES (one-equation subgrid-scale model)	GH	1.0	1	Pollutant concentration	–2.4–0
[82]	LES (one-equation subgrid-scale model)	GH, BC	1.0	1	Pollutant retention time,	–0.11 ~ 0.35
[60]	RNG $k-\epsilon$ model; Standard wall function	GH, UH, DH, AH	0.5–2.0	N/A	Air temperature	–1.07 ~ –0.06
[36]	RNG $k-\epsilon$ model; Standard wall function	GH, UH, DH	1.0	1–2	Pollutant concentration	–0.91 ~ –6.86
[27]	LES (Smagorinsky model)	GH, UH, DH	1.0	N/A	Exchange velocity,	–2.14–0
[35]	LES (Smagorinsky model)	GH, UH, DH	1.0	N/A	Sensible Heat Flux	–2.14–0
[34]	LES (one-equation subgrid-scale model)	GH	1.0, 2.0, and 0.5	1–2	Heat Flux, Pollutant Flux	–2.9 ~ –0.5
[83]	LES (turbulent kinetic energy)	NH, GH, UH, DH	1.0	1–3	Pollutant flux	–2.7–0
[84]	RNG $k-\epsilon$ model; Standard wall function	GH, UH, DH	1.0	1–3	Exchange velocities	–2.56 ~ –0.293
[85]	LES (Smagorinsky model)	NH, GH, UH, DH	0.5, 1.0 and 2.0	1–5	Pollutant concentration	–0.2
[86]	LES (one-equation subgrid-scale model)	GH, GC (Ground cooling)	1.0	1	Pollutant flux	–0.1, 0, 0.1, 0.188
[71]	realizable $k-\epsilon$ model; Near wall treatment	GH	0.5, 1.0 and 2.0	1–3	Air exchange rate, Retention time	–20–0
[87]	RNG $k-\epsilon$ model; Standard wall function	AH: All wall heated	0.5, 1.0 and 2.0	1–3	Predicted Mean Vote PMV index	N/A
[88]	RNG $k-\epsilon$ model; enhanced wall treatment	GH, UH, DH	3.0, 5.0	2–4	Age of air, Purging flow rate, Net escape velocity	–2.62, –4.36

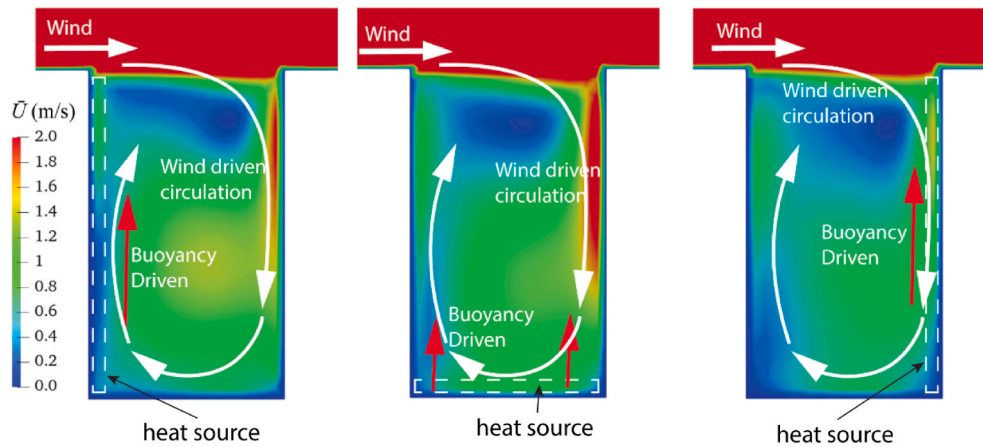


Fig. 2. Mechanism of the interaction between the wind-driven circulation and buoyancy force, from [28]. The dashed white box marks the location of heat emission. The white arrows represent the wind-driven flow direction and red arrows represent the buoyancy-driven direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

ignored in 2D street canyon, which may result in different air exchange rates [29]. Additionally, the spanwise airflow can be intensified by buoyancy. There is a lack of evidence to show multiple vortices flow structure still exist in a 3D street canyon. The computational cost for 3D simulation is much higher than 2D, over 100–1000 times, at the same grid resolution. Therefore, the wind-buoyancy ratio range in 3D simulation is narrower than in 2D, especially for the Ri number. There are few studies on the flow patterns under a wide range of aspect ratios. Additionally, the 3D flow pattern is difficult to characterize, as the vortices rotate in multiple planes [10]. As a result, the application of buoyancy using 3D simulations in urban design practice and parameterization is still limited.

2.3. Pure buoyancy-driven urban flow

The buoyancy plays a more important role in urban ventilation when there is a lack of ambient wind. The wind tunnel measurement [30] found that there is no vortices flow inside the street canyon when $Ri < -0.79$. Instead, the vertical plume flows dominate the street canyon. Under an extremely low wind scenario, the heated surfaces will generate thermal plumes under that condition, with warm air moving upwardly [108–110]. The closely spaced thermal plumes attract each other and generally merged into a single plume [52,111,112]. A series of water tank experiments were conducted at Hong Kong University to investigate the airflow induced by thermal buoyancy under calm conditions [113,114]. Unsteady vortices were observed in these experiments. They suggested that these vortices were caused by the interaction between thermal plumes generated by adjacent buildings. Fan et al. [115] summarized the studies on the thermal plume merging. They emphasized the development of dome or plume is largely determined by the size of the built-up area. Numerical simulations show that horizontal flow, which is generated by thermal plumes merging, also drives vortices flow inside the street canyon [29,116]. The air exchange velocity was found directly related to the wall temperature in these simulations.

This mechanism is supported by many field measurements. The field measurement conducted under a light and variable wind condition shows the in-canyon air temperature is lower than that above the rooftop, suggesting the existence of the thermal plumes merging between buildings [117]. The measurement in Athens shows that the air above the roof was 1.9°C hotter than the air below rooftop under a very weak wind scenario [24]. Another measurement in Athens also shows that the air above rooftop is about $3 \sim 5^\circ\text{C}$ hotter than the pedestrian level [118]. In that measurement, the airflow direction below the rooftop was found as opposed to the wind-driven flow, suggesting the buoyancy force dominates the flow. The field measurement conducted

in a high-rise building found the maximum wall-flow velocity was about $0.76 \sim 1.33 \text{ m/s}$ under very weak background wind [119]. A recent study indicates that the airflow at the pedestrian level under a pure buoyancy driven condition can reach about $1.5\text{--}4 \text{ m/s}$ in a real urban neighborhood [52]. This level of wind speed can be regarded as an effective source for urban ventilation.

3. Turbulence models

3.1. Measurement data for model validation

Before applying the turbulence model in urban ventilation analysis, it is necessary to validate it first through cross-comparing with measurement. The high-quality wind/water tunnel measurement is an important approach in validating CFD models. Table 3 summarizes the data availability of these measurements, which will help future CFD simulations to select suitable measurement data for cross-comparison and validation.

The non-isothermal wind/water tunnel experiments have been rarely conducted compared to the isothermal experiments due to their complicity. The pioneer non-isothermal wind tunnel measurement conducted by Uehara et al. [72] is the most used data for model validation, which provides the velocity and temperature profiles. However, Uehara's wind tunnel is limited with weakly unstable stratifications. As a result, numerical studies validated by this experiment are also limited to weakly unstable stratifications. Although 3D blocks were used, Uehara's experiment is similar to 2D experiments and only provides vertical flow measurement. There is a lack of measurement on flow and temperature distribution in horizontal directions. As a result, many subsequent 2D CFD simulations were conducted for model validation. The 3D effect is scarcely examined in these numerical studies.

In the past few years, the number of wind tunnel experiments has abruptly increased as the importance of urban buoyancy effect is aware. The use of non-intrusive measurements, e.g., Particle Image Velocimetry (PIV) and Laser Doppler Anemometer (LDA), has significantly improved the measurement quality and efficiency at the non-isothermal experiment. These measurements provide a wider range of Ri numbers and detailed information on flow, temperature, and pollutants, which could be very useful for validation studies. We suggest future studies to choose appropriate benchmarks based on the Ri number, Re number, heating position, and the data availability.

3.2. Performance of turbulence models

There are two types of turbulence models (i.e., the Reynolds-

Table 2

CFD studies on the urban thermal driven urban ventilation with idealized building array configuration and Realistic city configuration.

References	Turbulence model	Surface condition	Building geometry	Ventilation indices
[89]	RNG k - ϵ model, Standard wall function	Uniform heated	Cubical building array	Pollutant concentration
[90]	RNG k - ϵ turbulence model; Standard wall functions	Uniform heated	Cubical building array	/
[91]	LES (SGS- TKE)	Uniform heated	Cubical building array	scalar flux
[92]	Dynamic LES model; Wall function proposed by Wang and Moin	Uniform heated	Cubical building array	/
[93]	Standard k - ϵ turbulence; Standard wall functions	Uniform heated	Cubical building array	Average air temperature
[94]	LES (SGS- TKE)	Realistic urban heating from building energy simulation	Cubical building array	/
[3]	Standard k - ϵ turbulence model; Standard wall functions	Realistic urban heating from building energy simulation and solar radiation simulation	Building arrays with different morphologies	/
[95]	Realizable k - ϵ turbulence model; Standard Wall Function	Realistic urban heating from solar radiation simulation	Cubical building array	/
[96]	Standard k - ϵ turbulence model; Standard wall functions	Realistic urban heating from building energy simulation	Building arrays with different morphologies	Air temperature field;
[41]	Realizable k - ϵ turbulence model; Standard wall functions	Realistic urban heating from solar radiation simulation	Realistic city configuration	/
[59]	LES; WMLES (Algebraic Wall-Modelled LES Model)	Realistic urban heating from solar radiation simulation	Cubical building array	Air exchange rate
[97]	RNG k - ϵ turbulence model; Standard wall functions	Uniform roof cooling	Cubical building array	Air temperature; Pollutant concentration
[98]	Realizable k - ϵ turbulence model; Standard wall functions	Realistic urban heating from building energy simulation	Building arrays with different height	Heat fluxes
[99]	Realizable k - ϵ turbulence,	Realistic urban	Realistic city configuration	

Table 2 (continued)

References	Turbulence model	Surface condition	Building geometry	Ventilation indices
	Standard wall functions	heating from building energy simulation		Temperature field and wind velocity
[100]	Standard k - ϵ model; standard wall functions	Calculated from radiation heat transfer and moisture evaporation	3D street canyon	Average street-canyon air temperature
[101]	Realizable k - ϵ turbulence model; standard wall functions	Volumetric cooling power of roof greening	3D street canyon	Pollutant concentration
[102]	LES (SGS- TKE)	Realistic urban heating from building energy simulation	Cubical building array	Air exchange rate, Pollutant concentration
[103]	LES (Smagorinsky model)	Uniform bottom heating	Cubical building array	Airflow rate, Pollutant flux
[104]	Standard k - ϵ model	Uniform bottom heating	Cubical building array	Pollutant flow rate, air flow rate
[105]	Standard k - ϵ model	Realistic urban heating from building energy simulation	Cubical building array	Air change rate (ACH)
[106]	LES	Uniform heated	Cubical building array	Retention time
[107]	RNG k - ϵ model	Realistic urban heating with radiation simulation	3D street canyon	Air temperature, Pollutant concentration

averaged Navier–Stokes (RANS) models and Large Eddy Simulation (LES) models) to be applied to model non-isothermal airflow. Currently, a large number of turbulence models are validated by cross-comparing with the wind tunnel measurement [72], which provides data for Ri numbers ranging from -0.21 (unstable) to 0.78 (stable). The validation studies show RANS models can provide satisfactory results in reproducing the velocity and temperature profiles [64,71]. However, there are also significant differences in accuracy between various RANS models [52,116].

The wall functions that deal with the near-wall area with low Reynolds number were found of significant impact on the accuracy of RANS models. The Low Reynolds number modelling (LRNM) was found of higher accuracy than the standard wall functions [135]. The shortcoming of LRNM is the requirement of high density near wall mesh, which is not feasible for the full scale simulation. Customized wall functions [120,136] for buoyancy urban flow were found of good performance without using dense near wall mesh [137]. The combination of turbulence model and wall function was also found important for the modelling accuracy. The realizable k - ϵ model with LRNM and standard k - ϵ model with the NEWFs (Non-Equilibrium Wall Functions) were found of good performance [124]. Apart from wall functions, the grid structure also significantly affects the accuracy of non-isothermal modelling. Angelidis, Assimakopoulos and Bergeles [121] suggested that the local grid refinement can significantly improve RANS accuracy, while computational cells would not be significantly increased. A limitation of Uehara's wind tunnel experiments is that only the bottom surfaces were heated to weakly unstable stratifications.

Under the windward wall heating condition, LES models were found

Table 3

Wind/water tunnel experiments designed for investigating the buoyancy effect on urban ventilation. The Laser Doppler Anemometer (LDA), Laser Doppler Velocimetry (LDV), Hot-wire anemometer, and Particle Image Velocimetry (PIV) are used for velocity measurement. The Flame Ionization Detector (FID) and SF6 quantitative detector are used for pollutant concentration measurement. The thermocouples and Infrared thermography and cold-wire probe are used for temperature measurement.

Reference	Experiment platform	Ri number	Re number	Measurement method	Data availability	Previous validation references
[72]	Wind tunnel	-0.21–0.78	3500	LDA; Cold wire probe;	Velocity profile; Temperature profile; TKE profile;	[86,92,106,107, 120–122] and etc.,
[73]	Wind tunnel	-1.53~–0.058	9000–30,700	PIV; Thermocouples	Velocity profile; TKE profile;	[123–126]
[75]	Wind tunnel	-4.77–0	$4.32 \times 10^3 \sim 6.49 \times 10^4$	PIV; Thermocouples; SF6 quantitative detector;	Temperature contours; Velocity profiles;	[105]
[37]	Wind tunnel	-3.3 ~ -0.045	$4.28 \times 10^3 \sim 2.26 \times 10^4$	PIV; Infrared thermography	Temperature profiles; Pollutant concentration profiles;	N.A.
[103]	Wind tunnel	-0.15	5400	FID, LDV	Velocity fields; Temperature fields;	[103]
[113, 114]	Water tank	No wind		PIV	Mean and deviation profiles for velocity temperature and pollutant centration. Velocity profiles;	[52]
[127]	Wind tunnel	-1.18 ~ -1.27	$5 \times 10^3 \sim$	LDA; FID; fast-response cold-wire probe;	Velocity field; TKE field; temperature field; Turbulent heat flux; Concentration field;	N.A.
[128]	Wind tunnel	-0.28–0	5×10^3	LDA; FID; Fast-response cold-wire probe;	Velocity field; TKE field; temperature field; Turbulent heat flux; Concentration field;	N.A.
[129]	Wind tunnel	-2.94 ~ -1.61	5,700–7,200	PIV; FID;	Wind velocity profile; Pollutant concentration profile; TKE profile	N.A.
[130]	Wind tunnel	-0.13	5.98×10^4	Hot wire/film anemometer; SF6 quantitative detector;	Velocity profile; concentration profiles;	[130]
[131]	Wind tunnel	-0.23–0.29	$2.1 \times 10^4 \sim 3.2 \times 10^4$	Hot-wire anemometer; thermocouple; flame ionization detector;	Wind velocity; Air temperature; Air pollutant concentration;	N.A.
[132]	Wind tunnel	-1.09–0	3,040–25,300	Infrared thermography; PIV	TKE field; momentum flux field; temperature field; velocity profiles;	N.A.
[133]	Wind tunnel	-9 ~ -1.76	2.5×10^4	3D hot-wire anemometer; Thermocouples;	Air temperatures profiles; Velocity profiles;	[133]
[134]	Wind tunnel	-1.176–0.131	3189–9566	PIV, Thermocouples	Air temperature field; Velocity field;	N.A.
[30]	Wind tunnel	-2.07–0	6900	PIV	Flow field; Vorticity field	N.A.

of much higher accuracy than RANS models [123]. The interaction of buoyancy and wind-driven flow is the most intense under that condition, generating the multi-vortices structure [27]. Buoyancy also changes the turbulence coherent structure [30,86], which can be reproduced by LES, but not RANS models. Under the weak wind and buoyancy dominant conditions, only LES models show an acceptable accuracy [52]. However, it should be noted that RANS models show good performance under weak buoyancy and bottom heating conditions when appropriate computational meshes and wall functions were adopted. Therefore, RANS models are still good alternatives, with the validation by the aforementioned wind/water tunnel experiments. Given RANS models require much lower computational cost than LES models, more studies are worth to be conducted to investigate the sensitivity of RANS models on Ri .

4. Non-uniform wall surface temperature

Most of the existing studies summarized in Tables 1 and 2 applied the uniform wall surface temperature to represent the urban surface heated by solar radiation. However, wall surface temperature in urban areas is not uniform mainly due to the mutual shading between buildings. The non-uniform wall surface temperature may significantly affect the buoyancy effect, and thus change airflow and exchange velocity [59]. However, wall surface temperature simulation is a very challenging task for CFD as it is related to multiple physical processes, including solar radiation, longwave radiation, convection, evapotranspiration, and conduction to building materials. Currently, there are three approaches to include surface temperature models in urban natural ventilation

simulations, including multi-physics CFD simulation, surface temperature models, and Building Energy Simulation (BES)-CFD coupling method.

4.1. Multi-physics CFD models

Multi-physics features could be directly solved by CFD, as many CFD tools contain modules for both airflow, radiation, and conduction. The heat conduction, radiation, and convection are solved simultaneously, which makes the modelling intrinsically two-way coupling. Table 4 summarizes CFD studies, in which multi-physics models were applied. Some simulation results were validated by cross-comparing with field measurements, e.g. [41,138], and the validation results show multi-physics CFD models have acceptable accuracy in predicting wind speed and temperature fields. However, multi-physics CFD models have several shortcomings, which significantly limit their usage. Firstly, there is a lack of the longwave radiation model to include the impact of the sky into the model. Constant values of temperature at the top boundary have been generally used to represent the longwave radiative heat exchange [59,95]. Secondly, the dimension of building wall thickness is not comparable to the CFD computational domain, and makes it very challenging to generate the mesh for solid walls. Many studies used one-dimensional heat transfer model to avoid implicitly solving the heat transfer within building walls. Fourthly, building energy consumption, which significantly affects surface temperature distributions at the indoor side, is hard to be coupled into multi-physics CFD models. Last but not least, time scales for urban airflow and heat conduction at solid materials, e.g., building walls and soil, are significantly different. Due to

Table 4
Non-uniform wall temperature modelled with multi-physics CFD models. Turbulence model WMLES refers to Wall-Modelled subgrid-scale model and LES refers to Large Eddy Simulation. The Solar calculator is a tool in ANSYS FLUENT, which considers a beam using the position of the sun at any time during a year and applies a radiative heat flux at the wall type boundaries that are outside the shadow. The heat conduction model "Shell conduction" provides a convenient method to determine heat conduction in walls where the thickness is small to the overall geometry.

	Software	Incoming short-wave radiation	Long-wave radiation exchange	Radiation model	Wall model	Soil model	Turbulence model	Time step	Simulation Time
[41]	ANSYS FLUENT	Solar calculator	N/A	P1 radiation model	One-dimensional conduction shell conduction	10 m soil layer	Realizable k- ϵ model	15 min	5 days
[138]	ANSYS FLUENT	Solar calculator	N/A	P1 radiation model	One-dimensional conduction shell conduction	10 m soil layer	Realizable k- ϵ model	1 h	96 h
[95]	ANSYS FLUENT	Solar calculator	Set sky temperature of 255 K	Solar ray tracing algorithm (SRTA)	One-dimensional conduction	1.4 m soil layer	Realizable k- ϵ model	0.1 s	24 h
[59]	ANSYS FLUENT	Solar calculator	Set sky temperature of 255 K	Solar ray tracing algorithm (SRTA)	One-dimensional conduction	3.0 m soil layer	LES and Algebraic WMLES	0.1 s (Quasi-steady state)	1500s (Quasi-steady state)
[141]	ANSYS FLUENT	Solar calculator	N/A	P1 radiation model	One-dimensional conduction shell conduction	10 m soil layer	Realizable k- ϵ model	N/A	Steady-state simulation
[142]	ANSYS FLUENT	Solar calculator	N/A	P1 radiation model	One-dimensional conduction shell conduction	0.35 m soil layer	RNG k- ϵ model	N/A	Steady-state simulation
[143]	ANSYS FLUENT	Solar calculator	N/A	P1 radiation model	One-directional conduction	10 m soil layer	Realizable k- ϵ model	900 s	Five days
[144, 145]	Code_Saturne (open-source)	Measured meteorological radiation flux	Measured meteorological radiation flux	Discrete Ordinate Method (DOM)	One-dimensional conduction	Force-restore approach	Standard k- ϵ model	0.1 s-0.5 s	24 h

large heat capacity, the time span for heat conduction in solid materials could be several days, because heat conduction needs a long initial time to rule out the influence of initial value. However, the time step in the airflow simulation is about seconds to catch the turbulence variation and maintain the numerical stability. The above-mentioned factors require the CFD simulation to cover wide ranges at both spatial and temporal scales, and thus significantly increase the computational cost.

4.2. Urban surface temperature functions

An alternative approach to reduce the simulation cost is using wall temperature functions. [139,140]. developed a simulation code CSCRS (Coupled Simulations of Convection, Radiation, and Conduction), which includes a sub-model for wall temperature simulation. One shortcoming of CSCRC model is that the radiation model ignores the reflection of shortwave and longwave radiation. Additionally, the CSCRC code is not open source, which makes it hard to follow by other researchers. Recently, an urban surface temperature model (USM) was developed based on an open-source CFD code, PALM (Parallelized Large-Eddy Simulation Model, <https://palm.muk.uni-hannover.de/trac>) [20]. In USM, surface temperature is calculated separately, using surface temperature balance equation and multiple reflections of short and long wave radiation are resolved. These surface temperature models significantly reduce the requirement of computational resources. However, many simplifications and assumptions are still applied in USM model. For example, constant indoor temperature and homogeneous wall materials are still assumed at USM, which limit its accuracy.

4.3. BES-CFD coupling methods

Apart from CFD models, Building Energy Simulation (BES) models can also be used to calculate building surface temperature distributions by solving energy balance equations, including EnergyPlus [146], CitySim [147], TUF-IOBES [94], TUF-3D [45], and SOLNE [148]. BES models are suitable for wall surface temperature calculation because all the radiative transfer, heat conduction and storage in walls and soils, convective heat transfer, evapotranspiration, and building energy consumption are included in the modelling. Therefore, CFD simulation and BES models can work together efficiently to deal with the different physics at time scales in seconds and days respectively, and the scale compatibility problem can be solved [44,100,149,150]. On one hand, wall surface temperature or heat flux is calculated by BES models and transferred to CFD models as a boundary condition [94,96,99,105, 151–153]. On the other hand, CFD simulation results could also be used to improve the accuracy of BES in predicting surface CHTC (convective heat transfer coefficient). In a two-way coupling, the surface temperature will be transferred between BES and CFD multiple times to reach an equilibrium state. In the one-way coupling, only the surface temperature from BES model will be transferred to CFD model. Table 5 summarized studies focusing on coupling methods. Two-way coupling is essentially more accurate than one-way coupling, but as a tradeoff, two-way coupling is also more complicated and time-consuming. However, there is a lack of cross-comparison to show the improvement of prediction accuracy by two-way coupling. Multiple turbulence models are available in these CFD tools, including both LES and RANS models. This approach makes it possible for CFD simulations to balance accuracy and computational cost. The challenge of the BES-CFD coupling approach is the data transfer between two models.

5. Parameterization of the buoyancy effect on urban ventilation

The semi-empirical method is an alternative to the city scale modelling, since CFD simulation is too expensive [28,40,126]. The semi-empirical method does not directly model airflow and buoyancy but calculates geometrical and meteorological parameters and links the parameters with spatially averaged climate variables. Resolutions of

Table 5
Non-uniform wall temperature modelled by CFD-BEM coupling methods.

Reference	CFD model	Turbulence model	BES model	Coupling method
[102, 151]	PALM	LES	TUF-IOBES	One-way coupling: Surface heat flux
[94]	PALM	LES	TUF-IOBES	One-way coupling: Surface heat fluxes
[98,99]	OpenFOAM	Realizable k- ϵ model	CitySim	One-way coupling: Surface temperatures
[152]	OpenFOAM	N.A.	EnergyPlus	One-way coupling: Surface heat fluxes
[105]	STAR-CCM +	Standard k- ϵ model	CitySim	One-way coupling: Surface temperatures
[153]	STAR-CCM +	Standard k- ϵ model	TUF3D	One-way coupling: Surface heat fluxes
[96]	OpenFOAM	Standard k- ϵ model	CitySim	One-way coupling: Surface temperatures
[149]	Code-Saturne	Standard k- ϵ model	SOLENE	Two-way coupling: Surface heat fluxes
[150]	ANSYS FLUENT	Standard k- ϵ model	SOLENE	Two-way coupling: Surface temperatures
[100]	OpenFOAM	Standard k- ϵ model	BE-HAM model	Two-way coupling: Surface temperatures
[44]	ANSYS FLUENT	Standard k- ϵ model	BE-HAM model	Two-way coupling: Surface temperatures
[154]	OpenFOAM	Standard k- ϵ model	DCEP	One-way coupling: Surface temperatures
[155]	OpenFOAM	Standard k- ϵ model	EnergyPlus	One-way coupling: Surface temperatures
[156, 157]	OpenFOAM	Realizable k- ϵ model	ILLITHERM	One-way coupling: Surface temperatures
[158]	STAR-CCM+	standard k- ϵ model	CitySim	One-way coupling: Surface temperatures
[159]	Homemade program	Durbin model	1D heat conduction	One-way coupling: Surface temperatures

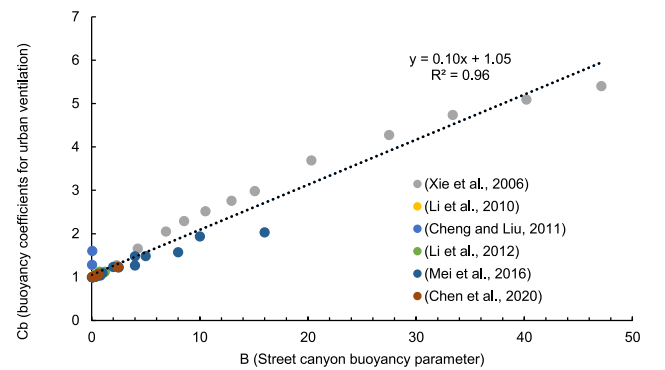
semi-empirical models are usually not as high as CFD simulation, but still good enough to support the decision-making at the city scale planning exercise [22].

To model the non-isothermal scenarios, semi-empirical models need to parameterize the buoyancy effect, which can be accomplished by linking geometrical parameters and the buoyancy effect modelled in CFD simulations or wind/water tunnel experiments. The mass and momentum exchange rate between air volumes below and above the roof level attracts the most attention. It is because both dilution rates of heat and pollutants are strongly related to the air exchange process, which has a direct impact on the urban thermal comfort and air quality [35,42,70]. The buoyancy coefficient (C_b) was applied to quantify the improvement of air exchange velocity caused by the buoyancy effect [28]. By applying the buoyancy coefficient, semi-empirical models at the city-scale under the isothermal condition, e.g. [28,40,49], can be extended to a more realistic non-isothermal condition. We defined the buoyancy coefficient (C_b) as the ratio between ventilation indices between isothermal condition and non-isothermal condition as [28]:

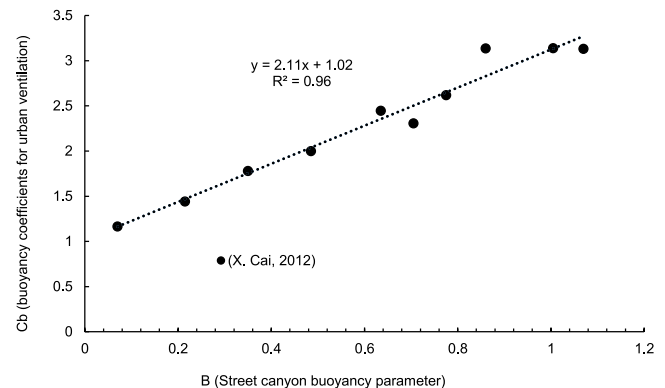
$$C_b = \frac{UE_t}{UE_n} \quad (5)$$

where UE_t is the exchange velocity for the non-isothermal condition and UE_n for isothermal conditions. Here C_b represents the air exchange rate improvement by buoyancy effect.

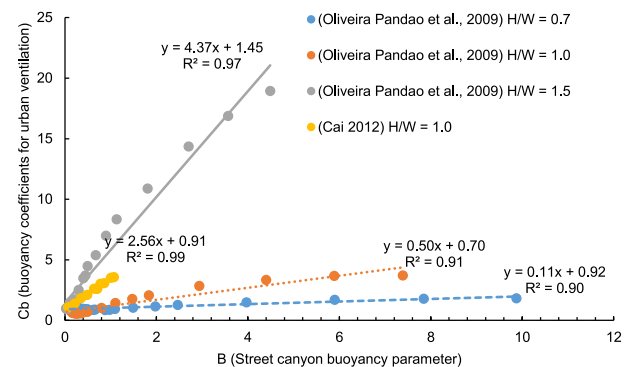
Using simple parameters to calculate the buoyancy coefficient is the reason why semi-empirical models are more feasible than CFD and wind/water tunnel experiments at the city scale. In this literature review, values of buoyancy coefficient for urban ventilation (C_b) were calculated based on the data from previous studies with heat sources at different locations. A linear fitting between the street canyon buoyancy parameter B and buoyancy coefficient C_b was conducted and the results are shown in Fig. 3. Dallman et al. [66] defined ΔT as the temperature difference between upwind wall and downwind wall. In the numerical simulation practice, unheated walls were generally assumed to be the



(a) Heated Ground (Street) Surfaces



(b) Heated upwind Facades



(c) Heated downwind Facades

Fig. 3. Correlations between buoyancy parameter B and buoyancy coefficient C_b inside a street canyon with the heat source at different locations, i.e., a: ground surfaces, b: downwind facades, and c: upwind facades. The buoyancy parameters and coefficients were calculated from previous studies of microscale numerical simulations that are listed in Tables 1 and 2

same temperature as ambient air or adiabatic. Therefore, the downwind wall temperature can be replaced by air temperature. In the present study, ΔT is calculated by the temperature difference between heated wall and ambient air. Then, B can be calculated for the upwind wall heating and ground heating.

For the ground heating condition (Fig. 3a), the root mean square error (RMSE) value is very close to 1, suggesting a strong linear correlation between C_b and B . Although multiple aspect ratios (H/W) are included, a linear correlation is well fitted in all aspect ratios. This is because the buoyancy parameter B is a combination of the Richardson number Ri and H/W . A strong linear relationship was also found between C_b and B for the upwind façade heating condition. The original B was derived from an upwind wall heating scenario, where the circulation driven by wind and buoyancy are of the same direction. Previous studies show, like upwind wall heating, ground heating also strengthens the vortex motion [34,83], i.e., wind and buoyancy driven airflow are of the same direction. This explains linear correlation between B and C_b in the ground heating and upwind conditions.

For the upwind façade heating condition, the linear relationship between C_b and B can still be found under the same aspect ratio at each individual study. Unlike the ground heating condition, the C_b varies greatly among different aspect ratios. For example, values of C_b with H/W 1.0 from [27] differs significantly from the ones [65] with the same aspect ratio. As mentioned in Section 2.2, the wind-driven and buoyancy-driven airflow are at opposite directions at the street canyon with the heated upwind facade, which may result in complex flow patterns with multiple vortices [11]. In the downwind wall heating cases, the buoyancy has a stronger impact in narrow street canyons. This is contrary to the definition of B , which shows wider street canyon has a stronger buoyancy impact. As suggested by Dallman [66], the buoyancy parameter B is only applicable to single vortex cell street canyon with assisting condition. In the downwind wall heating condition, buoyancy-driven flow is opposite to wind-driven flow, causing multiple vortices in the street canyon [11]. In that case, buoyancy plays a greater role as the wind-driven flow inside the deep street is weak.

The linear regression indicates that B and C_b are useful parameters to quantify the buoyancy impact of air exchange. However, the existing simulation data for calculating the buoyancy coefficient is still limited. More data with a wider range B is needed to parameterize buoyancy effect to improve the performance of semi-empirical models, especially for cities with complex 3D geometries. In a high-density complex 3D urban area, it is difficult to calculate the H/W , and would be more representative to parameterize buoyancy effect based on the 3D urban morphological parameters, such as frontal area density, site ratio, and height variation, in addition to the aspect ratio. These parameters better represent the urban geometry configuration and were already used in previous city scale models [22,28,48].

6. Conclusions and future directions

The evaluation of buoyancy effect on urban ventilation, especially on low wind conditions, is important for the extreme urban heat islands and pollutant islands prediction, and the future urban climate resilience analysis. This study reviewed recent research advances in buoyancy effect on urban ventilation, focusing on simulating and parameterizing the buoyancy effect on air exchange velocity. The main outputs are as follows:

- Calculating dimensionless numbers to evaluate the buoyancy effect is essential before conducting simulations for parametric studies. The Richardson number (Ri), Froude number (Fr), and street canyon buoyancy parameter (B) are recommended.
- The interaction between ambient wind and local buoyancy changes the airflow pattern in urban canopy layer. The buoyancy can cause a transition of street canyon flow from one-vortex to multi-vortex and significantly changes in air exchange velocity.

- The thermal plume merging is the main mechanism that drives the urban airflow under weak wind conditions. The airflow pattern, turbulence structure, and air exchange velocity are very different from wind-driven conditions.
- The isothermal wind tunnel experiments and their data availability are summarized. We suggest future simulations to use the newly emerged wind tunnel experiments with similar Richardson number and heating position as the performance of turbulence model related to them.
- Performance of turbulence models for non-isothermal simulation are summarized, and it is suggested to only use RANS models with weak buoyancy and heat sources at the ground of street canyons. Given RANS models require much lower computational cost than LES models, more studies are worth to be conducted to investigate the sensitivity of RANS models on Ri .
- Performance of surface temperature coupling methods is summarized to facilitate the multi-physics non-isothermal CFD simulation. The CFD-BES coupling method is recommended due to its high accuracy and acceptable computational cost.
- Last but not least, the parameterization of buoyancy effect on air exchange velocity is important for developing the city scale models with a strong buoyancy effect. A parameterization approach is proposed using the existing simulation data.

The current studies demonstrate the importance of buoyancy on urban ventilation. However, future studies are needed to provide practical suggestions for urban design and develop city-scale models. Research gaps in the existing studies are detected and summarized as below:

- The evaluation of buoyancy effect on urban ventilation strongly relies on the expensive measurements and CFD simulations. Once the geometry or temperature changes, new simulations and measurements are needed. The high cost limits its application in urban design practice. Future research on linking the air exchange velocity to the aspect ratio and dimensionless parameters is needed.
- Although several simulations used 3D building arrays to evaluate the buoyancy effect, a very narrow range of wind-buoyancy ratio and buildings forms were considered. The differences between 3D and 2D simulations are still not fully quantified. As a result, the implementation of extensive 2D street canyons simulation is limited, especially for high-density complex urban configurations.
- Several recent studies have investigated the airflow in the buoyancy dominant scenarios and indicate the airflow pattern is different from the wind-driven scenarios. However, the air exchange analysis under this scenario is not well understood. Future studies on heat and pollutant removal under buoyancy dominant conditions are needed.
- The database of air exchange velocity is still sparse and limited at a narrow range of buoyancy intensity. The parameterization of buoyancy effect on air exchange velocity for city scale needs more data for a wider range of wind-buoyancy ratio. To better integrate into the current city scale model, parameterization of buoyancy effect should be based on the 3D urban morphological parameters, such as frontal area density, site ratio, and height variation, in addition to the aspect ratio.

CRedit authorship contribution statement

Shuo-Jun Mei: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Chao Yuan:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is supported by Singapore Ministry of Education, NUS Presidential Young Professorship (Grant no.R-295-000-172-133) and Singapore National Research Foundation, Singapore ETH Centre, Future Resilience System II grant (Grant no. R-295-000-176-592).

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